Glossary

Simple definitions for Day 12 regression and classification terms

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Glossary

This glossary explains Day 12 terminology in simple language for quick revision and interview preparation.

Term	Simple definition
Alpha	Regularization strength. Higher alpha usually means stronger shrinkage.
AIC/BIC	Model comparison criteria that balance fit and complexity. Lower is usually better within comparable models.
Calibration	Whether predicted probabilities behave like real probabilities.
Coefficient	The model weight for a feature. It tells direction and strength of impact.
Confidence interval	Uncertainty range around an estimated average or coefficient.
Cook distance	A measure of how strongly one row influences the regression model.
ElasticNet	A model combining Ridge and Lasso penalties.
Extrapolation	Predicting outside the range seen in training data. Risky.
Gradient descent	An iterative method that reduces loss step by step.
Heteroscedasticity	Unequal residual spread across prediction levels.
Intercept	The predicted value when all features are zero.
Lasso	Regularized regression that can set weak coefficients to zero.
Leverage	How unusual a row is in the feature space.
Logistic Regression	A classification model that outputs probabilities.
Negative Binomial	A count model useful when counts are overdispersed.
OLS	Method that finds linear regression coefficients by minimizing squared residuals.
Overdispersion	When count variance is much larger than count mean.
Poisson Regression	A count regression model often used when mean and variance are close.
Prediction interval	Uncertainty range for one future observation.
Residual	Actual value minus predicted value.
Ridge	Regularized regression that shrinks coefficients but keeps features.
ROC-AUC	Ranking quality of a classifier across thresholds.
Sigmoid	Function that converts any score into a probability between 0 and 1.
VIF	Variance Inflation Factor; checks multicollinearity risk.

References and source foundation

The documents use the completed Day 12 notebooks as the main project evidence. The theory explanations are also aligned with the uploaded course materials and official library documentation.

Source	How it was used
Uploaded course guides: leit03.pdf and Haupt_leitfaden03.pdf	OLS, least squares, residuals, regression line, correlation, interpolation/extrapolation.
Uploaded ML slides: Methode der Kleinsten Quadrate and Typen des Machinellen Lernens	Supervised learning, regression vs classification, gradient descent, logistic regression context.
scikit-learn Linear Models User Guide - https://scikit-learn.org/stable/modules/linear_model.html	Linear Regression, Ridge, Lasso, ElasticNet, Logistic Regression, PoissonRegressor.
scikit-learn Model Evaluation User Guide - https://scikit-learn.org/stable/modules/model_evaluation.html	MSE, MAE, R2, ROC-AUC, PR-AUC, precision, recall, F1, calibration.
statsmodels GLM documentation - https://www.statsmodels.org/stable/glm.html	Generalized Linear Models, Poisson family and model interpretation.
statsmodels NegativeBinomial documentation - https://www.statsmodels.org/stable/generated/statsmodels.discrete.discrete_model.NegativeBinomial.html	Negative Binomial implementation details and count-model context.
UCLA IDRE Negative Binomial Regression notes - https://stats.oarc.ucla.edu/r/dae/negative-binomial-regression/	Practical explanation of Negative Binomial as a model for overdispersed count outcomes.